

Section L
Antiderivatives

$$1. f(x) = \int x^8 dx = \frac{1}{9} x^9 + C$$

$$2. f(x) = \int x^{-6} dx = -\frac{1}{5} x^{-5} + C$$

$$3. f(x) = \int x^{5/7} dx = \frac{7}{12} x^{12/7} + C$$

$$4. f(x) = \int \frac{4}{\sqrt{x}} dx = (4)(2)\sqrt{x} + C$$

$$5. f(x) = \int \frac{1}{2x^3} dx = \left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)x^{-2} + C$$

$$6. f(x) = \int x^3 \sqrt{x} dx = \int x^{7/2} dx \\ = \frac{2}{9} x^{9/2} + C$$

$$7. f(x) = \int \sqrt[3]{t^2} dt = \int t^{2/3} dt \\ = \frac{3}{5} t^{5/3} + C$$

$$8. f(x) = \int (x^3 - 2x + 7) dx \\ = \frac{1}{4} x^4 - (2)\left(\frac{1}{2}\right)x^2 + 7x + C$$

$$9. f(x) = \int \left(x^{-3} + x^{1/2} - 3x^{1/4} + x^2\right) dx \\ = -\frac{1}{2} x^{-2} + \frac{2}{3} x^{3/2} - (3)\left(\frac{4}{5}\right)x^{5/4} + \frac{1}{3} x^3 + C$$

$$10. f(u) = \int \left(u^{2/3} - 4u^{1/5} + 4\right) du \\ = \frac{3}{5} u^{5/3} - (4)\left(\frac{5}{6}\right)u^{6/5} + 4u + C$$

$$11. f(x) = \int \left(\frac{7}{x^{3/4}} - \sqrt[3]{x} + 4\sqrt{x}\right) dx \\ = (7)(4)x^{1/4} - \frac{3}{4} x^{4/3} + (4)\left(\frac{2}{3}\right)x^{3/2} + C$$

$$12. f(x) = \int (x + x^4) dx \\ = \frac{1}{2} x^2 + \frac{1}{5} x^5 + C$$

$$13. f(t) = \int (2 - t + 2t^2 - t^3) dt \\ = 2t - \frac{1}{2} t^2 + (2)\left(\frac{1}{3}\right)t^3 - \frac{1}{4} t^4 + C$$

$$14. f(x) = \int x^{1/3} (4 - 4x + x^2) dx \\ = \int \left(4x^{1/3} - 4x^{4/3} + x^{7/3}\right) dx \\ = (4)\left(\frac{3}{4}\right)x^{4/3} - (4)\left(\frac{3}{7}\right)x^{7/3} + \left(\frac{3}{10}\right)x^{10/3} + C$$

$$15. f(x) = \int \left(\frac{1}{x^2} - \cos x\right) dx \\ = -x^{-1} - \sin x + C$$

$$16. f(\theta) = \int (4 \sin \theta + 2 \cos \theta) d\theta \\ = -4 \cos \theta + 2 \sin \theta + C$$

$$17. f(y) = \int e^y dy = e^y + C$$

$$18. f(x) = \int x e^{x^2} dx \\ = \frac{1}{2} e^{x^2} + C$$

$$19. f(x) = \int \frac{2}{x} dx \\ = 2 \ln |x| + C$$